Identifying High-Value Locations and Best-Selling Products

An Analysis of Coffee Shop Sales Data

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1 Executive Summary

2 Introduction

The central problem addressed in this report is how a new café can choose a location and initial product offering that maximises early sales and customer engagement. The report focuses on two key objectives: identifying the location with the strongest potential for expansion, and determining which product should be prioritised at launch. Together, these findings will support strategic decision-making for launching a new café with the highest likelihood of early success.

2.1 Dataset Introduction

To tackle this problem, the analysis uses the Coffee Shop Sales dataset from Maven Analytics, which provides detailed transaction-level information across multiple existing stores. The dataset includes 149116 rows and 11 columns, representing a large volume of customer purchase activity. Key variables include numeric identifiers such as `transaction_id`, `store_id`, and `product_id`, which uniquely identify each transaction, store, and product. Temporal variables include `transaction_date` (stored as character) and `transaction_time` (stored as a time object), capturing when each purchase occurred. Measures of customer behaviour are represented through `transaction_qty`, a numeric variable indicating the number of items purchased per transaction. Categorical variables (stored as character) such as `store_location`, `product_category`, `product_type`, and `product_detail` describe where the transaction took place and the type of product purchased. Together, these variables provide a rich foundation for analysing sales patterns across locations and product categories.

3 Methodology

3.1 Executive Summary

3.2 Introduction

The central problem addressed in this report is how a new café can choose a location and initial product offering that maximises early sales and customer engagement. The report focuses on two key objectives: identifying the location with the strongest potential for expansion, and determining which product should be prioritised at launch. Together, these findings will support strategic decision-making for launching a new café with the highest likelihood of early success.

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3.4 Methodology

Before the data is used for analysis, a data cleaning process is performed. All variables are checked for missing values and no missing values are found across the dataset, thus no further action was taken. Next, the class of each variable is verified. The **transaction_date** variable, which contains date information, was stored as a character string and was therefore mutated to a date format to allow for further analysis.

Following the goal of determining the best location and best selling products, the relationship between product category, product details, sales quantity, unit price, and store location is explored.

3.5 Summary of data - Sales by location

Table 1: Total sales and quantity sold by store location

Store Location	Quantity Sold	Average Price	Total Sales
Hell's Kitchen	71737	3.30	236511.2
Astoria	70991	3.27	232243.9
Lower Manhattan	71742	3.21	230057.2

Table 1 presents the total quantity sold, average product price, and total sales of stores in each location. The store in Hell's Kitchen recorded the highest total sales amount among the three locations, despite having a slightly lower sales quantity compared to the store in Lower Manhattan. Nevertheless, the slightly higher average price contributed to the higher total sales figure.

3.6 Summary of data - Sales by product category

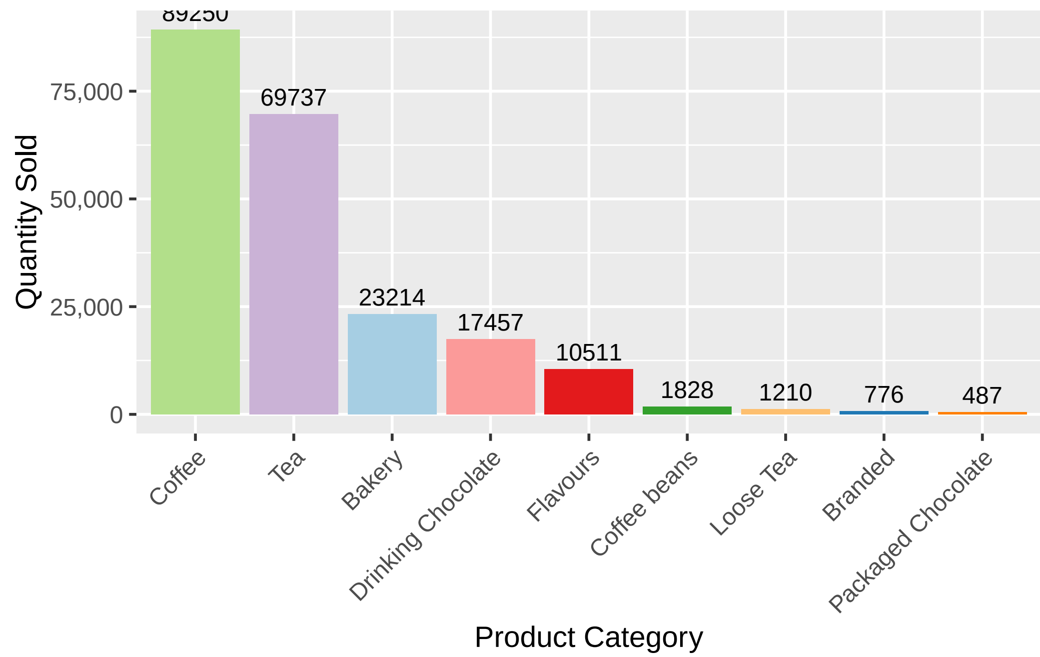

Figure 1: Sales by product category

Figure 1 presents the quantity sold by product category. Coffee is the best selling product category, followed by tea. The third best selling category is bakery items; however, the difference between tea and bakery sales is significant, as bakery sold less than half of total tea sold, representing a large gap between the top two best selling categories and the rest.

4 Results

4.1 Which Location would be Most Suitable based on Sales?

4.1.1 Overall Store Performance

Table 2: Summary statistics by store location in New York City.

Store Location	Total Sales	Total Transactions	Total Quantity Sold	Average Transaction Value	Average Daily Sales
Hell's Kitchen	236511.2	50735	71737	4.661696	1306.692

Table 2: Summary statistics by store location in New York City.

Store Location	Total Sales	Total Transactions	Total Quantity Sold	Average Transaction Value	Average Daily Sales
Astoria	232243.9	50599	70991	4.589891	1283.116
Lower Manhattan	230057.2	47782	71742	4.814726	1271.035

Table 2 presents the overall performance metrics across all three NYC locations for the six-month period.

1. *Hell's Kitchen* led on total sales ($236,511), total transactions (50,735), and average daily sales ($1,306.69), suggesting it attracts the highest volume of customers consistently.
2. *Astoria* came in second, recording $232,244 in total sales and an average daily sales figure of $1,283.12.
3. *Lower Manhattan* ranked third in total sales ($230,057) and had the fewest transactions (47,782), but has the highest average transaction value at $4.81 (which is interestingly higher than that of Hell's Kitchen ($4.66) and Astoria ($4.59)). This implies that while Lower Manhattan sees less foot traffic, customers there tend to spend more per visit, potentially suggesting more premium drinks or a higher-income demographic.

Across all three locations, the differences in total sales are narrow - spanning just ~$6,500 from top to bottom - indicating a broadly competitive and fairly matched market.

Similarly, total quantity sold was nearly identical between Hell's Kitchen (71,737) and Lower Manhattan (71,742) despite Lower Manhattan having ~3,000 fewer transactions, further supporting the pattern of higher spend-per-visit in that location.

4.1.2 Total Sales by Store Location in NYC

Figure 2 shows the total sales across the six-month period are similar across all three locations, with **Hell's Kitchen leading, followed by Astoria and Lower Manhattan**.

4.1.3 Monthly Sales Trends

Figure 3 reveals a **consistent and strong upward trajectory in sales** across all three locations from January to June. All stores dipped slightly in February before recovering and rising gradually to June. This seasonal period likely reflects increased foot traffic in the warmer months, where customers are more likely to leave their homes and visit coffee shops.

By June, Hell's Kitchen pulled ahead with the highest monthly sales (~$57,000), followed by Astoria (~$55,500) and Lower Manhattan (~$54,500). The convergence of all three lines

Figure 2

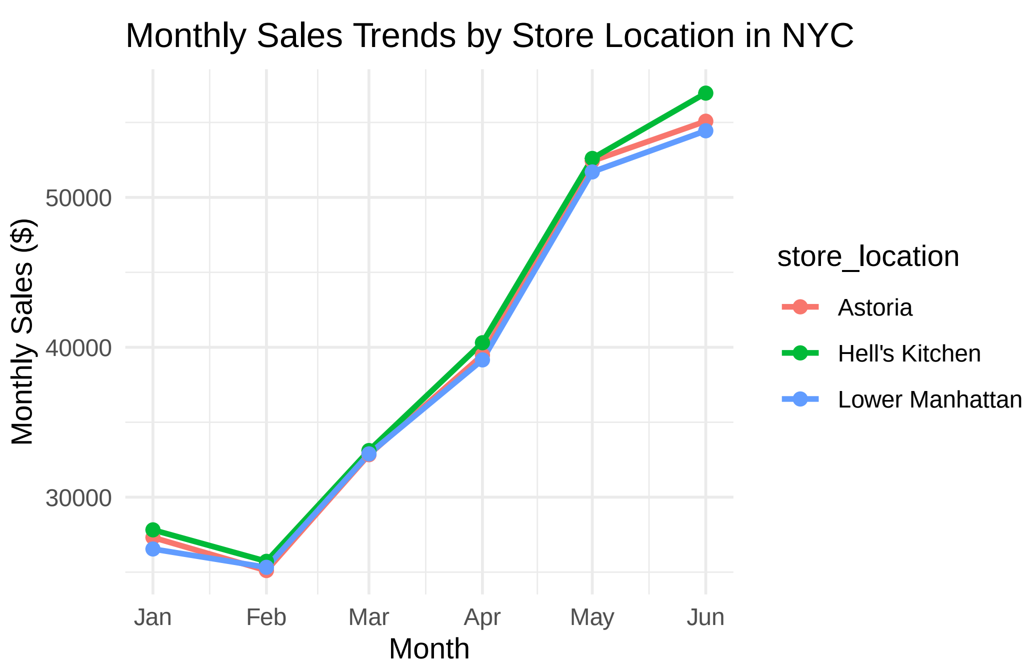

Figure 3

through March and April indicates similar demand between all three locations during these months, while the slight divergence in May to June suggests **Hell's Kitchen has a modest but growing edge heading into summer**.

4.1.4 Peak-Hour Demand Analysis

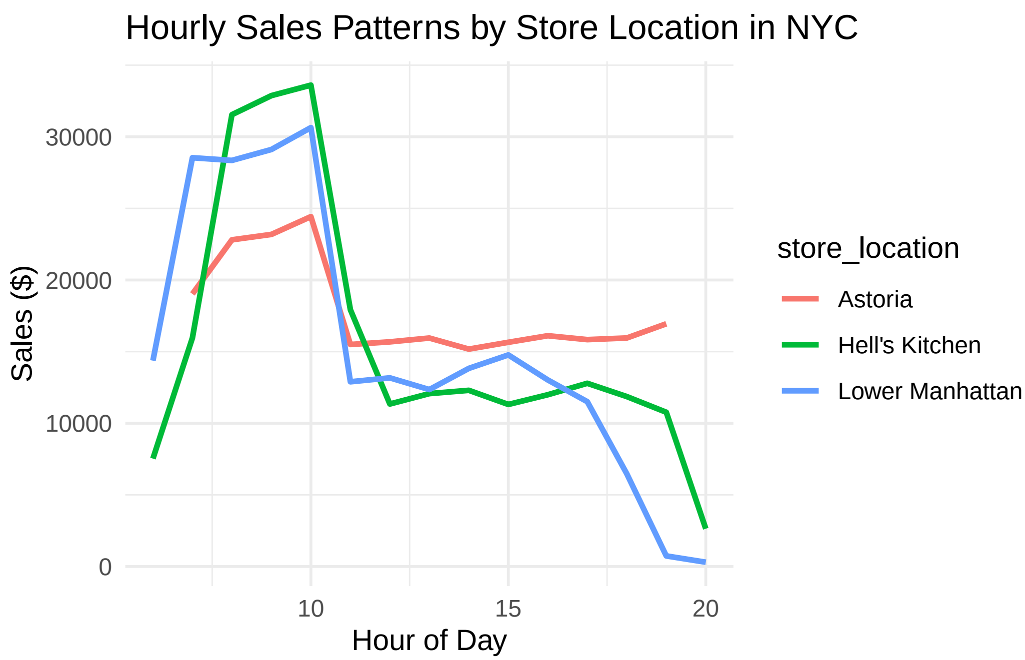

Figure 4

As shown in Figure 4, **all three locations experience the same morning peak**, with sales increasing between 7 to 10 AM. Hell's Kitchen and Lower Manhattan both reached peaks exceeding $30,000 in sales during this time period, aligning with the high foot traffic from morning commuters. After 10 AM, Hell's Kitchen and Lower Manhattan decline steeply, while Astoria maintains a more stable sales curve throughout the day, with sales ranging over $15,000 to $17,000, suggesting a more consistent, neighbourhood-driven customer base rather than commuter-dependent.

Lower Manhattan drops to nearly $0 by 8 PM, and Hell's Kitchen follows a similar pattern, indicating that these locations are correlated with commuter and office traffic. On the other hand, Astoria sustains sales throughout the evening.

4.1.5 Conclusion

This analysis examined the sales performance of Hell's Kitchen, Astoria, and Lower Manhattan, across total sales, transaction behaviour, monthly trends, and hourly patterns.

All three locations performed similarly in aggregate, with total sales differing by less than 3% over the six-month period. Monthly trends were consistent across locations, with a dip in February followed by sustained growth through to June, indicating that seasonal demand, rather than location-specific factors, is the dominant driver of revenue fluctuations.

However, there are meaningful differences to note: Hell's Kitchen and Lower Manhattan both exhibited sharp morning peaks, with sales concentrated heavily in the 7 to 10 AM window and declining steeply after, pointing to a strong dependence on commuter and office-worker traffic. Astoria, on the other hand, maintained a flatter and more stable sales curve throughout the day, suggesting a loyal neighbourhood-based customer base that generates consistent demand across multiple days.

Taking all findings together, **Astoria is recommended as the strongest candidate for a new café**. Even though Hell's Kitchen seems the best when it comes to raw total sales and daily averages, its revenue is heavily peak-dependent, which introduces operational risk during off-peak hours. Astoria's more evenly distributed demand, competitive overall sales, and neighbourhood-driven patronage provide a more stable and sustainable commercial foundation, which are vital for a new café seeking to establish a consistent customer base from the start.

4.2 Optimal Menu Recommendations

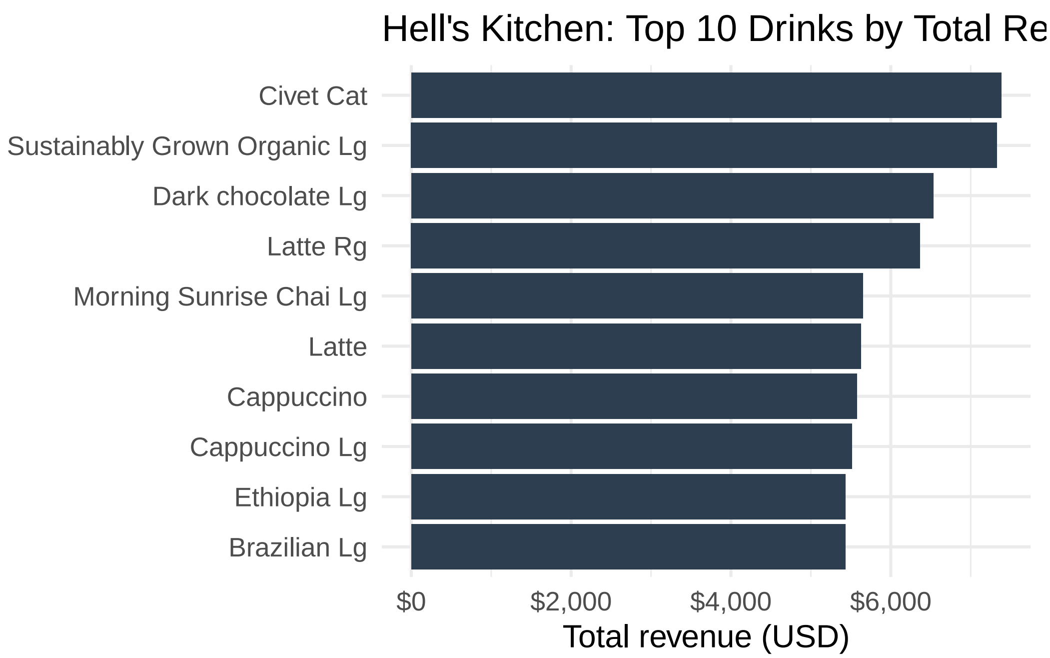

Figure 5: top 10 drinks by total revenue (hell's kitchen)

As shown in Figure 5, total revenue in Hell's Kitchen is concentrated among a small set of drinks, with a clear drop-off after the top-ranked items. The highest-earning products include premium coffees and larger-size beverages, suggesting that both higher unit prices and steady demand contribute to revenue. This figure highlights the key drivers of takings and provides an evidence-based starting point for deciding which drinks should be prioritised for menu placement and promotion.

Table 3: top 10 drinks by total revenue (hell's kitchen)

drink	total_quantity	total_revenue
Civet Cat	164	7380.00
Sustainably Grown Organic Lg	1543	7329.25
Dark chocolate Lg	1452	6534.00
Latte Rg	1498	6366.50
Morning Sunrise Chai Lg	1413	5652.00
Latte	1500	5625.00
Cappuccino	1487	5576.25
Cappuccino Lg	1297	5512.25

Table 3: top 10 drinks by total revenue (hell's kitchen)

drink	total_quantity	total_revenue
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00

Table 4: top 10 drinks by total quantity sold (hell's kitchen)

drink	total_quantity	total_revenue
Ouro Brasileiro shot	1854	5056.20
Serenity Green Tea Rg	1601	4002.50
Morning Sunrise Chai Rg	1589	3972.50
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00
Our Old Time Diner Blend Sm	1550	3100.00
Sustainably Grown Organic Lg	1543	7329.25
English Breakfast Lg	1529	4587.00
Earl Grey Rg	1522	3805.00
Brazilian Sm	1520	3344.00

Tables Table 3 and Table 4 together identify the best menu candidates by combining profitability and demand. Items near the top of both tables are strong "core menu" choices because they sell frequently and generate high total revenue, making them reliable for day-to-day operations. Drinks that rank highly in Table 3 but not Table 4 likely achieve revenue through a higher price point; these are good premium offerings to keep for margin and upselling, but they may require targeted promotion and careful stock planning. Conversely, drinks that are very popular in Table 4 but lower in Table 3 are dependable volume sellers and can be used as anchors for bundles or add-ons. Overall, a balanced menu should prioritise overlapping winners, then supplement with a small set of premium specials.

5 References

Free Sample Dataset Download - Coffee Shop Sales - Maven Analytics | Build Data Skills, Faster. https://mavenanalytics.io/data-playground/coffee-shop-sales. Accessed 21 May 2026.

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